Laboratory Work No. 1

Modeling of a Permanent-Magnet Synchronous Motor

Variant 3

Goal: to study the mathematical model of a PMSM and inspect its transient responses under sinusoidal stator excitation and a step load torque.

1. Mathematical Model

The simulated PMSM is described in the stationary alpha-beta frame by the standard current, speed, and position state equations from the assignment. The total flux is the sum of the stator inductive flux and the permanent-magnet flux. In this implementation the viscous friction coefficient was taken as B = 0 because it is not specified in the variant table.

Parameter	Value
Variant	3
L, H	0.033000
R, Ohm	5.3000
Jm, kg*m^2	0.000600
Pole pairs np	5
lambda_m, Wb	0.1600
B, N*m*s/rad	0.0000

2. Simulation Scheme

The numerical model reproduces the same signal flow as the Simulink implementation: voltage source -> PMSM dynamics -> signal scopes, with an external load torque step applied at 0.5 s.

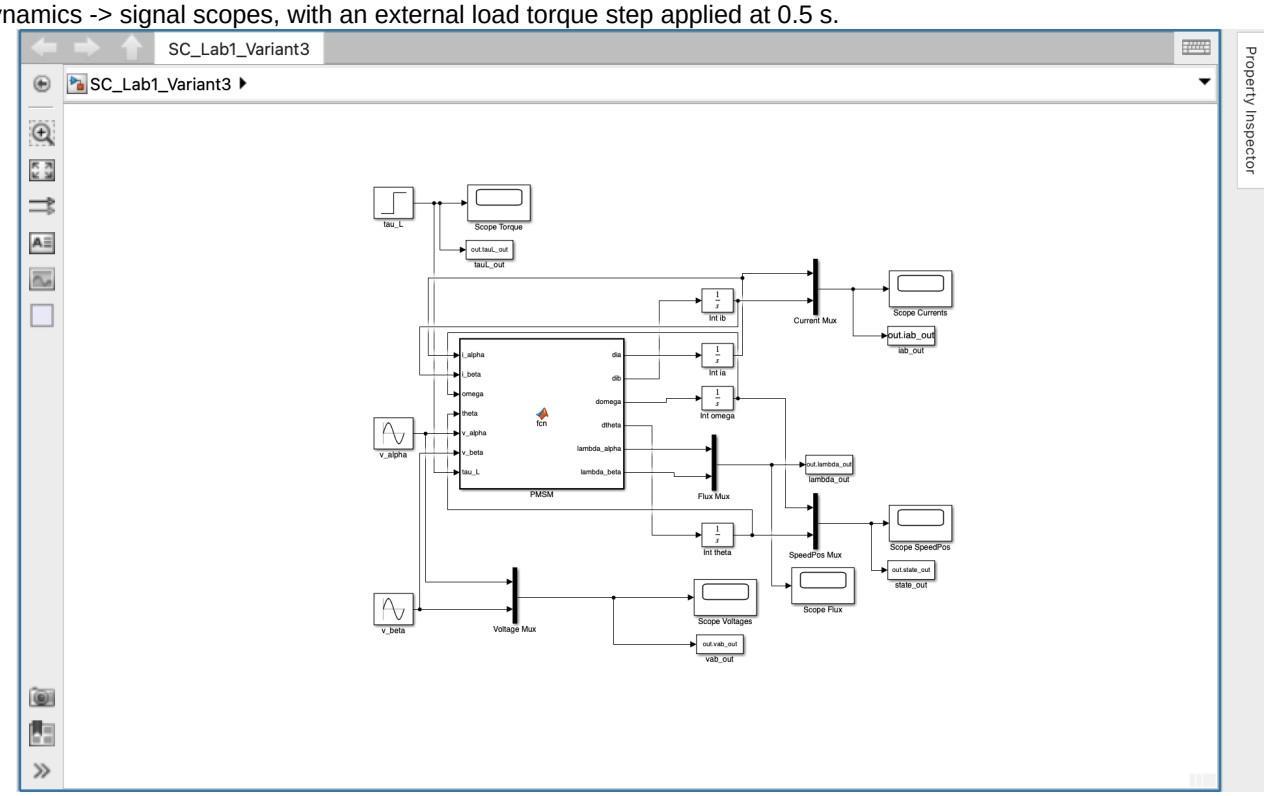

Figure 1. Main Simulink scheme of the PMSM model.

3. Simulink Screenshots and Transient Responses

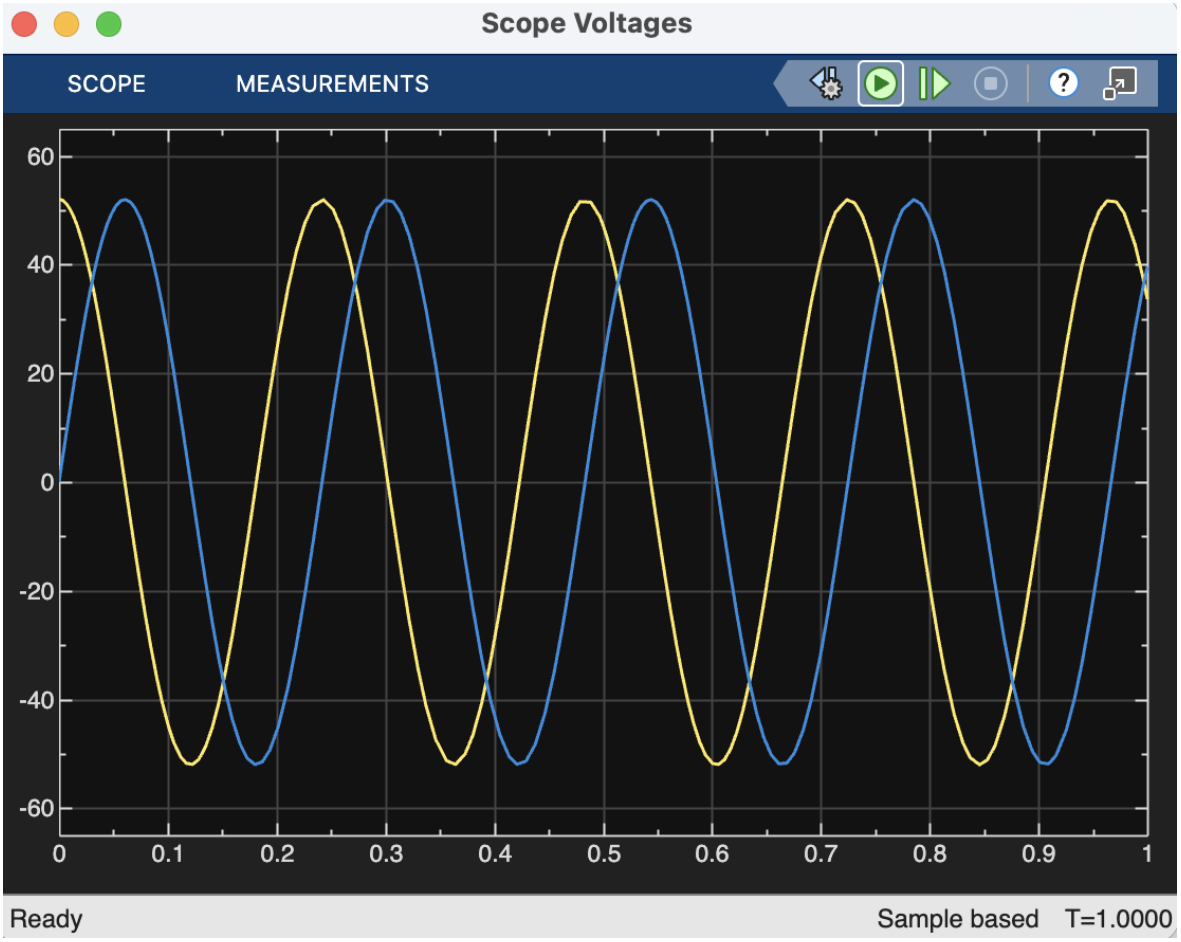

Figure 2. Scope of the applied stator voltages v_alpha and v_beta.

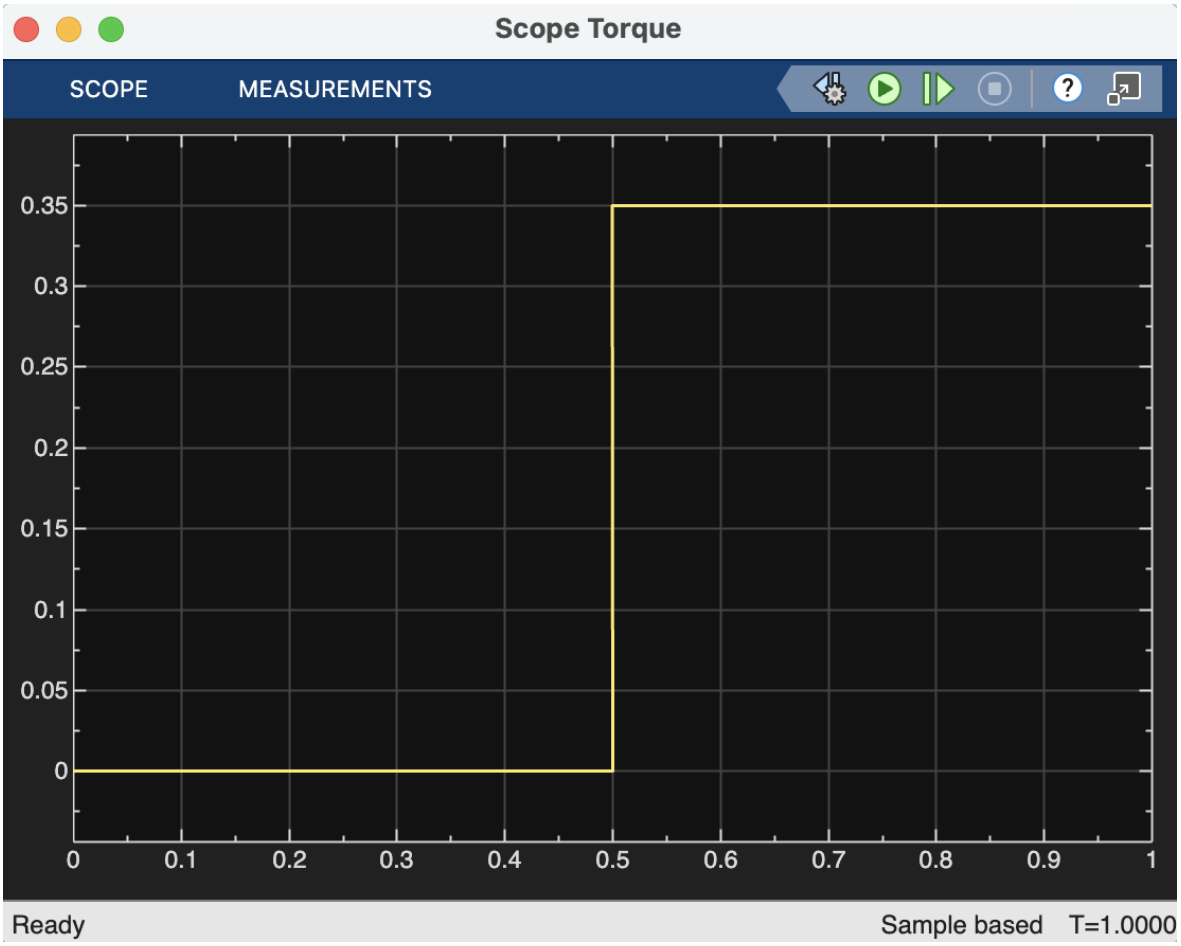

Figure 3. Scope of the load torque tau_L.

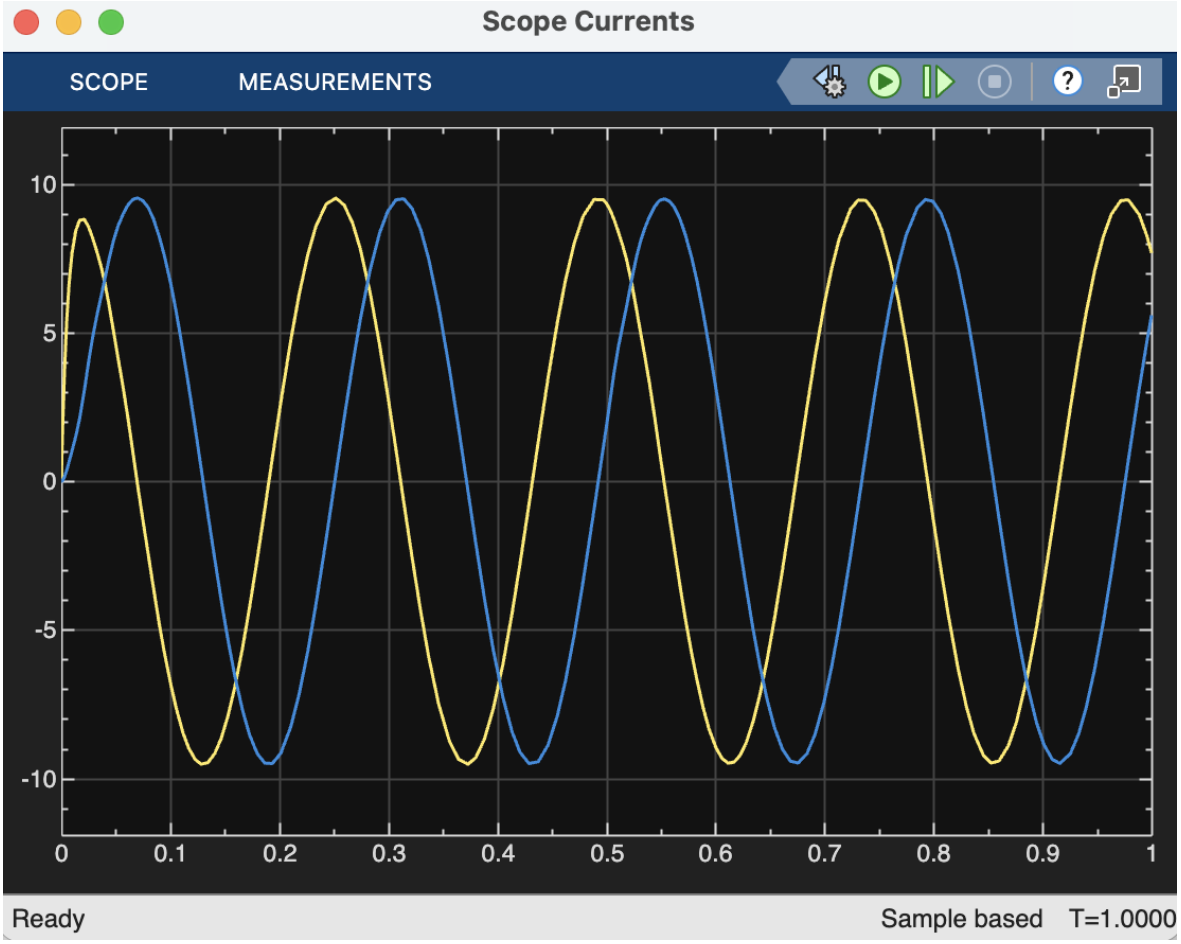

Figure 4. Scope of the stator currents i_alpha and i_beta.

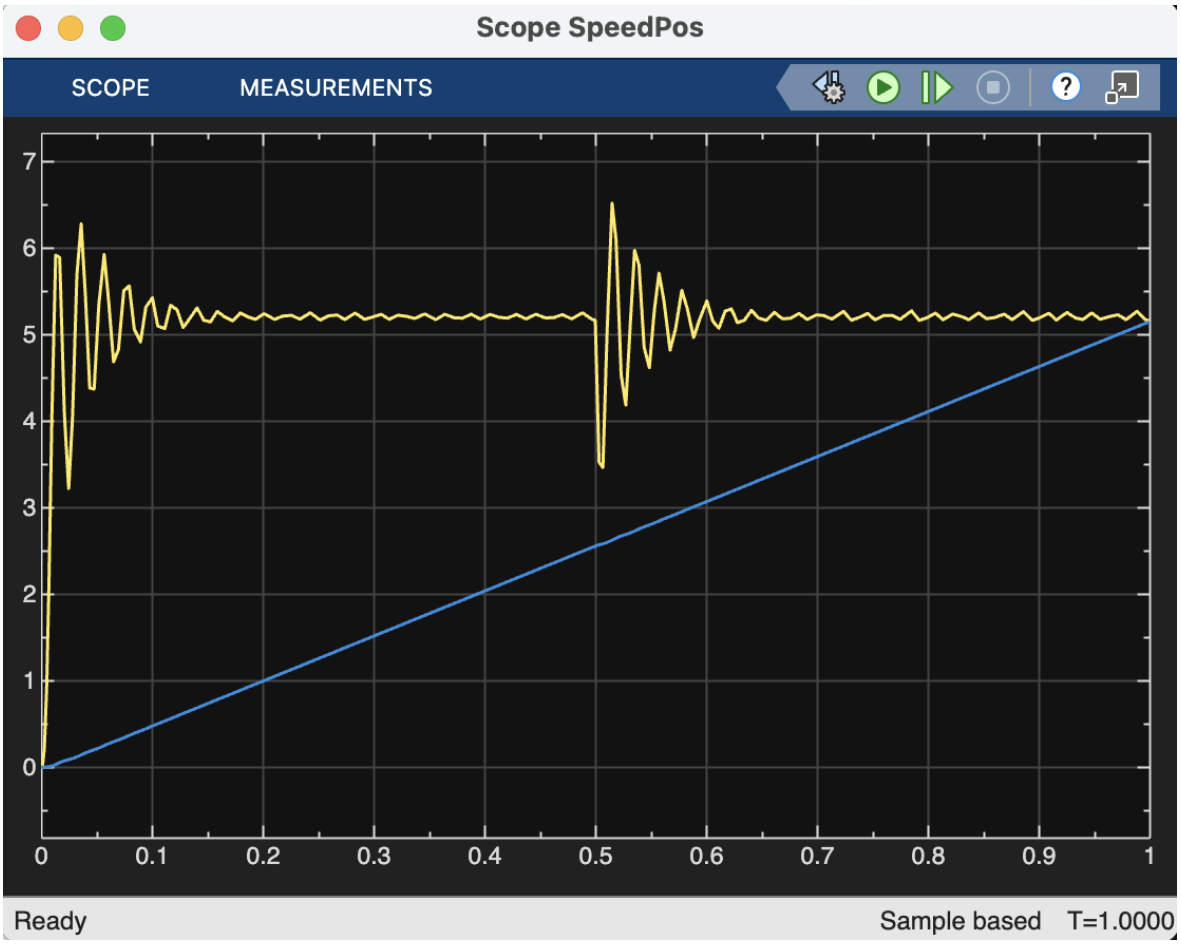

Figure 5. Scope of the rotor speed omega and rotor position theta.

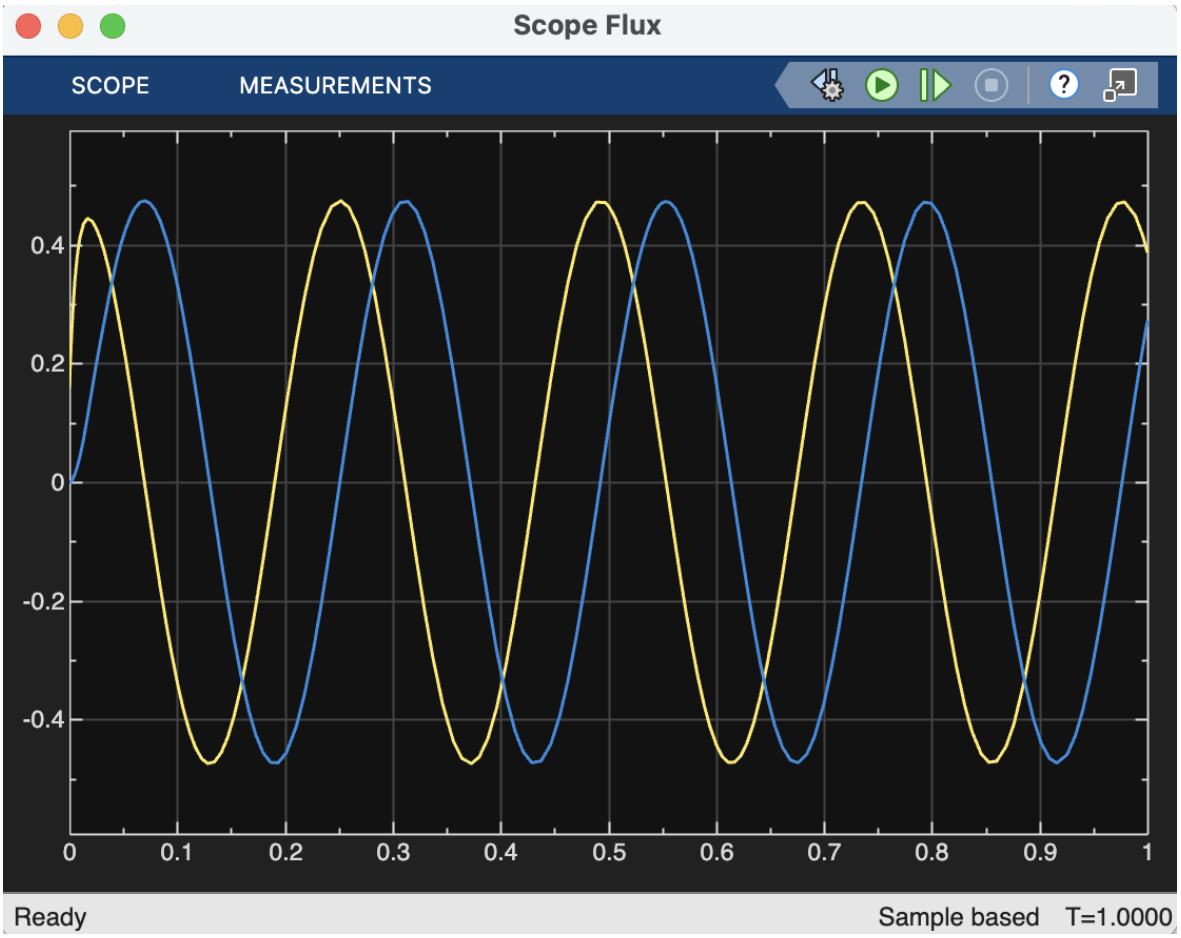

Figure 6. Scope of the total magnetic flux components lambda_alpha and lambda_beta.

4. Selected Values

Quantity	Value
lambda_alpha(0), Wb	0.160000
lambda_beta(0), Wb	0.000000
omega(0.4), rad/s	5.199866
omega(1), rad/s	5.200012
theta(1), rad	5.143008

5. Analysis and Conclusion

The model shows a smooth electromagnetic start-up under sinusoidal excitation. After the load-torque step at 0.5 s the speed slope decreases, while the rotor position keeps increasing monotonically. The flux components stay bounded and reflect the interaction between stator inductance and permanent-magnet flux.